

3.5 Topic 3: Biodiversity Under Threat

Biodiversity

Biodiversity is a key resource, which provides a range of valuable goods and critical services to human populations. Biodiversity results from natural physical processes, and as such has distinct geographical patterns. Locally, where biological resources are valued short-term for their immediate economic potential, biodiversity is often under threat through over-exploitation. However, global threats such as climate change and the role of alien species are also important.

Wellbeing

Increasingly it is recognised that human wellbeing and ecological wellbeing are inter-linked, and that biological resources need to be managed. However, concepts of the 'value' of these resources vary between different players and agreement is difficult to reach. There is a wide spectrum of management options, both locally and globally, each with its own merits and disadvantages. Reconciling the desire for development and the need to manage and maintain biodiversity is a key challenge for the future.

1 Defining biodiversity

Enquiry question: *What is the nature and value of biodiversity?*

What students need to learn	Suggested teaching and learning
■ There are different ways of defining biodiversity in terms of genetic, species and ecosystem diversity; each has its own merits.	■ Investigating and comparing ways in which biodiversity can be defined.
■ There are a range of key processes and factors that influence biodiversity; such as the role of endemism, climate and human activity and actions.	■ Investigating the processes that determine biodiversity, and their relative importance.
■ The global distribution of biodiversity and biodiversity hotspots reveal important patterns, and suggests that pivotal areas exist.	■ Recognising that biodiversity is unevenly spread and that some highly biodiverse, threatened areas can be described as 'hotspots'.

■ Ecosystems have value and importance in terms of biodiversity and ecological resources which should be illustrated with reference to a named global ecosystem (economic, cultural and environmental), in terms of the goods and services that they provide to different groups. This includes recognition that biodiversity is not equally valued by all.	■ Developing an appreciation of the value of biodiversity and using examples to illustrate this, and exploring the value of biodiversity and ecosystem resources to different interest groups.
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2 Biodiversity threats

Enquiry question: *What factors and processes threaten biodiversity?*

What students need to learn	Suggested teaching and learning
■ The distribution of threatened areas (hotspots under threat, and areas with threatened species) reveals global patterns.	■ Investigating the global pattern of threatened areas.
■ Global factors threaten biodiversity (from climate change and rising sea levels for instance) as well as local factors, which are often related to economic development and direct ecosystem exploitation, as well as attitudes to the value of biodiversity.	■ Developing an understanding of the location of threatened areas in terms of both local and global threats and their results.
■ Ecosystem processes (energy flow and nutrient cycles) can be disrupted by these threats, such as by the introduction of alien species or changes in nutrient levels.	■ Investigating how ecosystem processes can be disrupted by threats.
■ The link between economic development and ecosystem destruction/degradation is complex, and can be illustrated by examining named ecoregions – pristine areas, degraded areas and protected areas.	■ Weighing up and illustrating the strength and nature of the link between economic development and ecosystem degradation and conservation.

3 Managing biodiversity

Enquiry question: *Can the threats to biodiversity be successfully managed?*

What students need to learn	Suggested teaching and learning
■ The concept of sustainable yield is a way of determining the 'safe' use of ecosystems, and therefore the required balance between conservation, management and development.	■ Consideration of the utility of the concept of sustainable yield, as a way of determining the 'safe' use of an ecosystem in relation to development.
■ The role of different players (individuals, interest groups, international organisations) is important in managing biodiversity, as is the extent to which conflict between players can be resolved, if it can be resolved at all.	■ Weighing up and illustrating the roles of named players, and explaining why they often conflict with one another.
■ There is a spectrum of strategies and policies for managing biodiversity, each with advantages and disadvantages for different interest groups; these should be applied globally and to named examples of management.	■ Weighing up the costs and benefits of a variety of named strategies and policies, and illustrating these, for example international agreements, global frameworks, genetic banks, biosphere reserves, sustainable management, protection and restoration.
■ The future of biodiversity is uncertain; some attempts have been made to determine its future and its impacts, such as the MEA, but the desire to develop presents continuing global and local threats which are not easily resolved without making difficult choices.	■ Exploring a range of futures, and developing an appreciation of the considerable uncertainty of the biodiversity future, ranging from 'business as usual' to radical choices that might reduce the threat.